

ENGR 102 Summer Bridge

Day 15 Instructor Key

Append, Concatenation, Slicing, and Strings as Sequences

Friday, July 24, 2026 • 50 points

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Daily target

increasing independence

Process strings by index and build stored results with append, concatenation, and direct slice boundaries.

Allowed tools	Prior tools plus string indexing, in, direct slicing, list append, and list/string concatenation
Support supplied	The input/output contract and allowed sequence tools are supplied.
You must decide	Traversal state, collected-result state, slice boundaries, and validation order.
Scaffold level	Specification only; no planning prompts, variable inventory, or ordered algorithm.

Scoring and completion standard

50 points

Evidence	Points	Secure work shows
Integrated trace	9	intermediate state, condition results, and exact output
Diagnosis and repair	7	actual, intended, cause, repair, and a boundary test when requested
Full program and tests	34	coherent executable logic, required output, and defended tests

Independence standard. You may use the supplied requirements and examples, but you must produce one connected solution. A collection of disconnected correct lines is not yet a complete program.

Begin with the trace, then use its evidence habits when you construct.

Task 1. Integrated trace**9 points**

Trace index selection, append state, slicing, and concatenation in one sequence ledger.

```
labels = ["A", "B"]
text = "loop"

for index in range(1, len(text), 2):
    labels.append(text[index])

tail = text[1:3]
combined = labels + [tail]
print(labels)
print(tail)
print(combined)
```

Your task

1. Give the complete index/list/slice state and all exact output lines. Explain why the slice stop is not included.

Answer and explanation

1. The range indexes are 1 and 3, so text[1] is o and text[3] is p. labels becomes ['A', 'B', 'o', 'p']. The slice text[1:3] includes indexes 1 and 2 but excludes 3, so tail is oo. combined is ['A', 'B', 'o', 'p', 'oo']. These are the three printed lines in order.

Task 2. Diagnose and repair**7 points**

The intent is to create [1, 2, 3], but the program fails before print.

```
values = [1, 2]
values = values + 3
print(values)
```

Your task

1. Name the actual failure, explain the type mismatch, give two minimal repairs with different sequence operations, and name one empty-list test.

Answer and explanation

1. The + expression raises TypeError because list concatenation requires another list, not the integer 3. Repair 1: values = values + [3]. Repair 2: values.append(3). Starting from [] should produce [3] under either repaired pattern.

Task 3. Construct a complete program**34 points across Pages 4-5****Program specification**

- Read word as text. If its length is less than 4, print invalid and nothing else.
- Otherwise loop through every index and append characters at even indexes to even_characters.
- Count vowels using membership in the text aeiouAEIOU.
- Build edge_text by concatenating the first two-character slice and the last two-character slice.
- Print even_characters, vowel_count, and edge_text on separate lines.

Answer and explanation

```
word = input()

if len(word) < 4:
    print("invalid")
else:
    even_characters = []
    vowel_count = 0

    for index in range(len(word)):
        character = word[index]
        if index % 2 == 0:
            even_characters.append(character)
        if character in "aeiouAEIOU":
            vowel_count = vowel_count + 1

    edge_text = word[:2] + word[len(word) - 2:]
    print(even_characters)
    print(vowel_count)
    print(edge_text)
```

The validation branch protects both slices. The loop uses indexes because selection depends on index parity; membership tests the stored character value.

The first slice stops before index 2. The second slice starts at length minus 2 and runs through the end. String + concatenates the two slices.

Task 3 continued. Test and defend the program**included in 34 points****Expected tests and results**

1. sensor -> ['s', 'n', 'o'], vowel_count 2, edge_text seor.
2. code -> ['c', 'd'], vowel_count 2, edge_text code.
3. abc -> invalid only.

Scoring guidance

4 validation; 5 initialization; 6 full traversal; 5 even-index collection; 4 vowel count; 5 correct slices/concatenation; 2 output; 3 tests/explanation.

Accept alternative correct code when it obeys the stated tool boundary, handles the required cases, and produces the required output. All blue text is answer or scoring guidance.

VIP Lab Alignment

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This page adds a current lab handoff while preserving the workbook's independence-ramp tasks.

Why this page is here

VIP Lab Day(s): 6

Activities: 18.23, 18.24

Apply filtering and the same normalization pipeline to strings before counting or comparing.

Open these current notebook cells

Activity / stable cells	Notebook work	Evidence to carry forward
18.23 18-23-work / 18-23-transfer / 18-23-cold	Count Kept Code Characters	Traverse a string one character at a time.
18.24 18-24-work / 18-24-transfer / 18-24-cold	Badge ID Match	Apply the same cleanup rule to both strings.

Readiness check before you code

1. For Activity 18.23, name the characters that do not count and the condition that skips them.

Answer and explanation: Use the exact skip characters supplied in the cell and a 'not in' membership check so only kept characters increment the count.

2. For Activity 18.24, write the cleanup operations in the same order for both identifiers.

Answer and explanation: Both identifiers must remove spaces and underscores and normalize case before equality is evaluated. Applying different cleanup to the two sides invalidates the comparison.

Notebook and submission procedure

1. Use the sample and transfer cells for formative practice; attempt before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only the Studio cells listed above are scored for independent mastery in the matching Lab Day notebook.
3. Save or download the completed notebook as one .ipynb file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a .py file or create a ZIP.